

OOP 2 單位換算器

學號：A1123327

姓名：陳佳思

這次做單位換算器時，我先使用 AI 幫忙產生基本的程式架構。不過第一次的程式有一些地方有問題，所以不能直接執行。我後來自己測試程式，找到錯誤的地方並進行修改，最後讓單位切換和換算功能都可以正常使用。

 English   Copy

Title: Recreate a "Unit Converter".

Requirements:

- Window size: 480 × 280, using BorderLayout.
- NORTH: A JComboBox for selecting the conversion type (Length / Weight / Temperature).
- CENTER: Two rows. Each row contains one JTextField and one JComboBox (source unit and target unit).
- When the conversion type is changed, the options in both unit JComboBoxes should change accordingly.
- Length: Meter, Centimeter, Inch, Foot.
- Weight: Kilogram, Gram, Pound, Ounce.
- Temperature: Celsius, Fahrenheit, Kelvin.
- SOUTH: A "Convert" button. The result should be displayed in the second JTextField.

[Give feedback](#)

Java Code

```
// Java

import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class UnitConverter extends JFrame {

    private JComboBox<String> typeCombo;
    private JComboBox<String> sourceUnitCombo;
    private JComboBox<String> targetUnitCombo;

    private JTextField sourceField;
    private JTextField resultField;
```



```
// Unit lists
private String[] lengthUnits = {
    "Meter", "Centimeter", "Inch", "Foot"
};

private String[] weightUnits = {
    "Kilogram", "Gram", "Pound", "Ounce"
};

private String[] temperatureUnits = {
    "Celsius", "Fahrenheit", "Kelvin"
};

public UnitConverter() {

    // Window settings
    setTitle("Unit Converter");
    setSize(480, 280);
    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    setLayout(new BorderLayout());

    // -----
    // NORTH
    // -----

    typeCombo = new JComboBox<>{
        new String[]{"Length", "Weight", "Temperature"}
    };

    JPanel northPanel = new JPanel();
    northPanel.add(typeCombo);

    add(northPanel, BorderLayout.NORTH);

    // -----
    // CENTER
    // -----
```



How the GUI is organized



For example,

- ```

4 Series Length
5 Enter 100
6 Source = Cartesian
7 Target = Vector
8 Click Compute
9 Result = 1.0

```